Status	Finished
Started	Sunday, 18 October 2026, 12:02 PM
Completed	Sunday, 18 October 2026, 12:10 PM
Duration	7 mins 59 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is small and unsorted ✓
c. When the list is small and sorted
d. When the list is large and sorted

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94
b. 90 and 99 ✓
c. 89 and 94
d. 89 and 99

Question **3**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 500
c. 1000
d. 10 ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(log n) ✓
- b. O(n log n)
- c. O🥐
- d. O(1)

Question **5**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. At the beginning of the list
- b. In the middle of the list
- c. In the right sublist ✓
- d. In the left sublist

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. Item is the last element in the array or item is not there at all
- b. When the item is the last element in the array
- c. When the item is not the array at all
- d. When the item is somewhere in the middle of the array ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The last element is returned
- c. The first element is returned
- d. An error is raised

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Must use a sorted array

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Searching ✓

b. Mining

c. Discovering

d. Finding

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓

b. By using a hash function

c. By starting from the first element

d. By comparing each element sequentially

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Linear search ✓

b. Dynamic search

c. Binary search

d. Hashing

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It does not work with negative numbers
- c. It is slower than linear search for small lists
- d. It can only be applied to large lists

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By sorting the list first
- b. By dividing the list into halves
- c. By using a hash function
- d. By comparing each element in the list sequentially ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search stops
- c. The search continues in the left or right sublist ✓
- d. The search continues from the beginning

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search divides the list into halves.
- b. Linear search requires the list to be sorted.
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search can be applied to both sorted and unsorted lists. ✓